

AI for Biomedical Systems & Critical Systems

Safety, reliability, regulation, and deployment patterns

Kelly Powell

UAB Heersink School of Medicine

June 2026

Why this topic matters

AI is transforming biomedical engineering — from imaging and diagnostics to wearables, prosthetics, and personalized treatment planning. In critical systems, the same capabilities create higher stakes: errors can affect patient safety, infrastructure, or essential services.

- Biomedical AI must be accurate, explainable, and clinically useful.
- Critical AI must also be robust, monitored, and governable.
- The best systems combine performance with safety engineering.

Core use cases

- Medical imaging and computer-aided diagnosis.
- Predictive analytics for patient deterioration.
- Personalized medicine and treatment optimization.
- Smart devices, wearables, and prosthetics.

Value created

- Faster detection and triage.
- Better use of complex multi-modal data.
- More adaptive care pathways.
- Continuous monitoring outside the clinic.

What makes a system critical

A system is critical when failures can cause serious harm — especially to health, rights, or essential services. That changes the engineering bar from “works well” to “works reliably under uncertainty.”

- **Safety impact:** harm to patients or operators.
- **Operational impact:** outages, delays, or cascading failures.
- **Decision impact:** automated recommendations in high-stakes settings.
- **Governance impact:** auditability, accountability, and oversight.

A practical AI lifecycle for critical deployments

- 1 Define intended use, boundaries, and failure modes.
- 2 Validate data quality, representativeness, and provenance.
- 3 Test performance across subgroups and edge cases.
- 4 Deploy with human oversight, monitoring, and rollback paths.
- 5 Reassess drift, incidents, and post-market performance.

Safety and reliability controls

Technical controls

- Redundancy and failover.
- Input/output validation.
- Robustness and adversarial testing.
- Continuous monitoring for drift and anomalies.

Operational controls

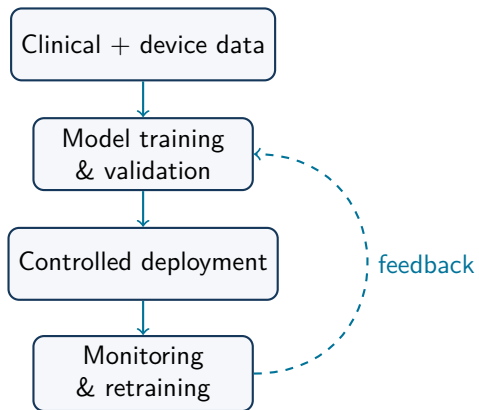
- Human-in-the-loop review for high-impact decisions.
- Clear escalation and incident response.
- Versioning and rollback capability.
- Periodic audits and documentation updates.

Governance framework: NIST AI RMF

The NIST AI Risk Management Framework organizes trustworthy AI work into four connected functions. It treats AI risk as an ongoing program rather than a one-time review.

Function	Purpose in critical biomedical AI
Govern	Assign accountability, policy, and oversight.
Map	Identify context, stakeholders, and harms.
Measure	Evaluate model performance, bias, and robustness.
Manage	Mitigate risks, monitor operations, and respond to incidents.

Deployment architecture



Research and practice agenda

- Build domain-specific benchmarks for biomedical reliability and safety.
- Combine model metrics with system-level metrics like latency and downtime.
- Study human-AI collaboration in clinical workflows.
- Treat cybersecurity, privacy, and safety as shared design requirements.
- Use post-deployment monitoring to close the loop from research to care.

Closing message

AI in biomedical systems is most valuable when it improves outcomes without introducing unacceptable risk. In critical systems, success depends on pairing machine learning with engineering discipline, governance, and continuous oversight.

Takeaway

High-performing AI is not enough; critical biomedical AI must also be safe, traceable, and resilient.

Selected references I

 UMass Library, *LaTeX: Beamer (Presentations)*, 2024.

 A. N. Bahkali et al., *Ten simple rules for effective presentation slides*, 2021.

 NIST, *AI Risk Management Framework (AI RMF 1.0)*, 2023.

 Public reviews and overviews on AI in biomedical engineering and biomedical sciences.